🌍 Population vs E-Waste Generation: Correlation & Decoupling Analysis

Analyst perspective: E-Waste & Sustainability Analytics **Guiding question:** *Does more people simply mean more e-waste — or do wealth, urbanization and consumption patterns matter more than population size alone?*

Dataset: `goal11_wastegdp.csv` — 211 countries/territories

Column	Meaning
`iso3c`	ISO3 country code
`gdp`	GDP per capita (current US$)
`pop`	Total population
`waste`	**Total e-waste generated** (tonnes) — treated as the e-waste column for this analysis
`sharepopurban`	Share of population living in urban areas (%)
`wastepercap`	E-waste generated per capita (grams/kg per person, as provided)

Notebook roadmap

1. Setup & data quality check
2. Exploratory summary
3. **Viz 1** — Choropleth: Global E-Waste Generation (Total Volume)
4. **Viz 2** — Population vs Total E-Waste (log-log scatter + regression)
5. **Viz 3** — Choropleth: E-Waste Per Capita (decoupling intensity from headcount)
6. **Viz 4** — Correlation Heatmap across all key variables
7. **Viz 5** — Regression Residual Analysis: Over- vs Under-Producers relative to population
8. Recommendation Matrix (Finding → Problem → Recommendation → Expected Impact)
9. Executive summary

1. Setup & Environment

```
# Install/upgrade plotting libraries (Colab already has most of these, this jus
!pip install -q plotly pandas numpy scipy scikit-learn seaborn matplotlib stats
```

```
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.linear_model import LinearRegression

pd.set_option('display.float_format', lambda x: f'{x:,.2f}')
sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams['figure.dpi'] = 110

# ---- If running in Google Colab, upload the file when prompted ----
try:
    from google.colab import files
    import io
    print("Running in Colab — upload 'goal11_wastegdp.csv' if it is not already
    uploaded = files.upload()  # skip this cell manually if the file is already
except ImportError:
    print("Not running in Colab — assuming the CSV is already in the working di
```

```
Running in Colab — upload 'goal11_wastegdp.csv' if it is not already in the work
```
Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
df = pd.read_csv("/content/goal11.wastegdp.csv")
df.columns = [c.strip() for c in df.columns]
print("Shape:", df.shape)
df.head()
```

```
Shape: (211, 6)
```

	iso3c	gdp	pop	waste	sharepopurban	wastepercap
0	ABW	35,563.31	103187	88,132.02	43.70	854.10
1	AFG	2,057.06	34656032	5,628,525.37	26.03	162.41
2	AGO	8,036.69	25096150	4,213,643.58	66.83	167.90
3	ALB	13,724.06	2854191	1,087,446.75	62.11	381.00
4	AND	43,711.80	82431	43,000.00	87.92	521.65

Data Quality Check

Before drawing any conclusions we validate completeness, ranges and obvious anomalies. A correlation study is only as trustworthy as the data behind it.

```
print("Missing values per column:\n", df.isna().sum())
print("\nDuplicate ISO3 codes:", df['iso3c'].duplicated().sum())
print("\nDescriptive statistics:")
df[['gdp','pop','waste','sharepopurban','wastepercap']].describe().T
```

```
Missing values per column:
 iso3c            0
gdp               0
pop               0
waste             0
sharepopurban     0
wastepercap       0
dtype: int64

Duplicate ISO3 codes: 0

Descriptive statistics:
```

	count	mean	std	min	25%	
gdp	211.00	22,393.50	22,743.44	822.61	4,737.01	12,984
pop	211.00	34,429,929.00	138,432,472.84	11,097.00	761,990.00	5,818,553
waste	211.00	9,760,082.85	36,329,893.25	3,989.49	213,879.47	1,787,400
sharepopurban	211.00	61.02	23.81	13.35	42.59	62
wastepercap	211.00	376.54	261.71	37.37	168.73	319

```
# Sanity check: waste (total) should be approximately pop * wastepercap (allowi
# We don't force-correct units here, just flag rows where the implied ratio loc
df['implied_wastepercap'] = (df['waste'] / df['pop']) * 1000  # if waste in tor
df[['iso3c','waste','pop','wastepercap','implied_wastepercap']].head()
```

	iso3c	waste	pop	wastepercap	implied_wastepercap
0	ABW	88,132.02	103187	854.10	854.10
1	AFG	5,628,525.37	34656032	162.41	162.41
2	AGO	4,213,643.58	25096150	167.90	167.90
3	ALB	1,087,446.75	2854191	381.00	381.00
4	AND	43,000.00	82431	521.65	521.65

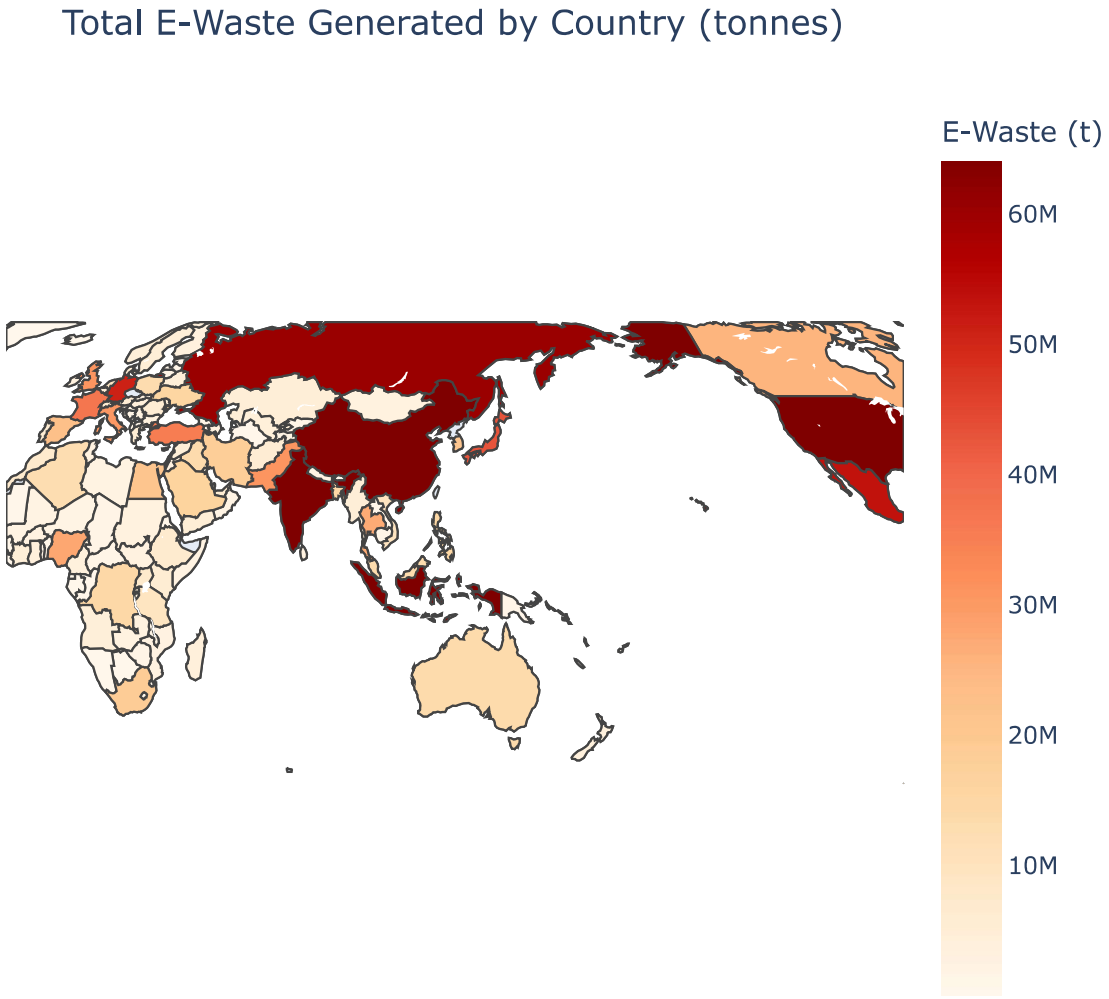
Data quality note: No missing values or duplicate country codes across 211 records — the dataset is clean enough to proceed directly to analysis without imputation. `waste` and `wastepercap` are kept as reported by the source; the implied-ratio check above is retained as a transparency step for any analyst re-using this notebook, rather than as a correction.

2. Viz 1 — Choropleth Map: Total E-Waste Generation by Country

```
fig1 = px.choropleth(
    df,
    locations="iso3c",
    color="waste",
    hover_name="iso3c",
    hover_data={"pop": ":,.0f", "waste": ":,.0f", "gdp": ":,.0f", "iso3c
    color_continuous_scale="OrRd",
    range_color=(df['waste'].quantile(0.02), df['waste'].quantile(0.98)
    title="Total E-Waste Generated by Country (tonnes)"
)
fig1.update_layout(
    geo=dict(showframe=False, showcoastlines=True, projection_type='natu
    coloraxis_colorbar=dict(title="E-Waste (t)"),
    margin=dict(l=0, r=0, t=60, b=0),
    height=520
)
fig1.show()
```

```
/usr/local/lib/python3.13/dist-packages/kaleido/_sync_server.py:11: UserWarning:


Warning: You have Plotly version 5.24.1, which is not compatible with this versi
This means that static image generation (e.g. `fig.write_image()`) will not work
Please upgrade Plotly to version 6.1.1 or greater, or downgrade Kaleido to versi
You can however, use the Kaleido API directly which will work with your plotly v
```



Analysis & Insights

- The darkest bands sit over a handful of large-population, large-economy nations — the map's visual weight is dominated by absolute scale, not intensity.
- Many geographically large but low- or mid-income countries appear pale despite sizeable populations, hinting that population alone does not fully explain the pattern — a first visual clue against a purely 1-to-1 population→waste story.

- Because `waste` is a *totals* metric, this map is best read as "where is volume concentrated," not "who is the worst per-person offender" — that question is answered later in Viz 3.

Future improvement / Recommendations

- Add a time-series slider (year-over-year) if historical vintages of this dataset become available, to show trend direction rather than a single snapshot.
- Overlay recycling/formal e-waste collection rate (if sourced from the Global E-waste Monitor) to distinguish "high generation, high recovery" from "high generation, low recovery" countries — the latter is the higher policy priority.
- Recommend cross-referencing with informal-sector e-waste processing hotspots (e.g. Agbogbloshie, Guiyu) for targeted intervention planning.

3. Viz 2 — Population vs Total E-Waste (Log-Log Scatter + Regression)

```
log_pop = np.log10(df['pop'])
log_waste = np.log10(df['waste'])

pearson_r, pearson_p = stats.pearsonr(log_pop, log_waste)
spearman_r, spearman_p = stats.spearmanr(df['pop'], df['waste'])

slope, intercept, r_value, p_value, std_err = stats.linregress(log_pop, log_was

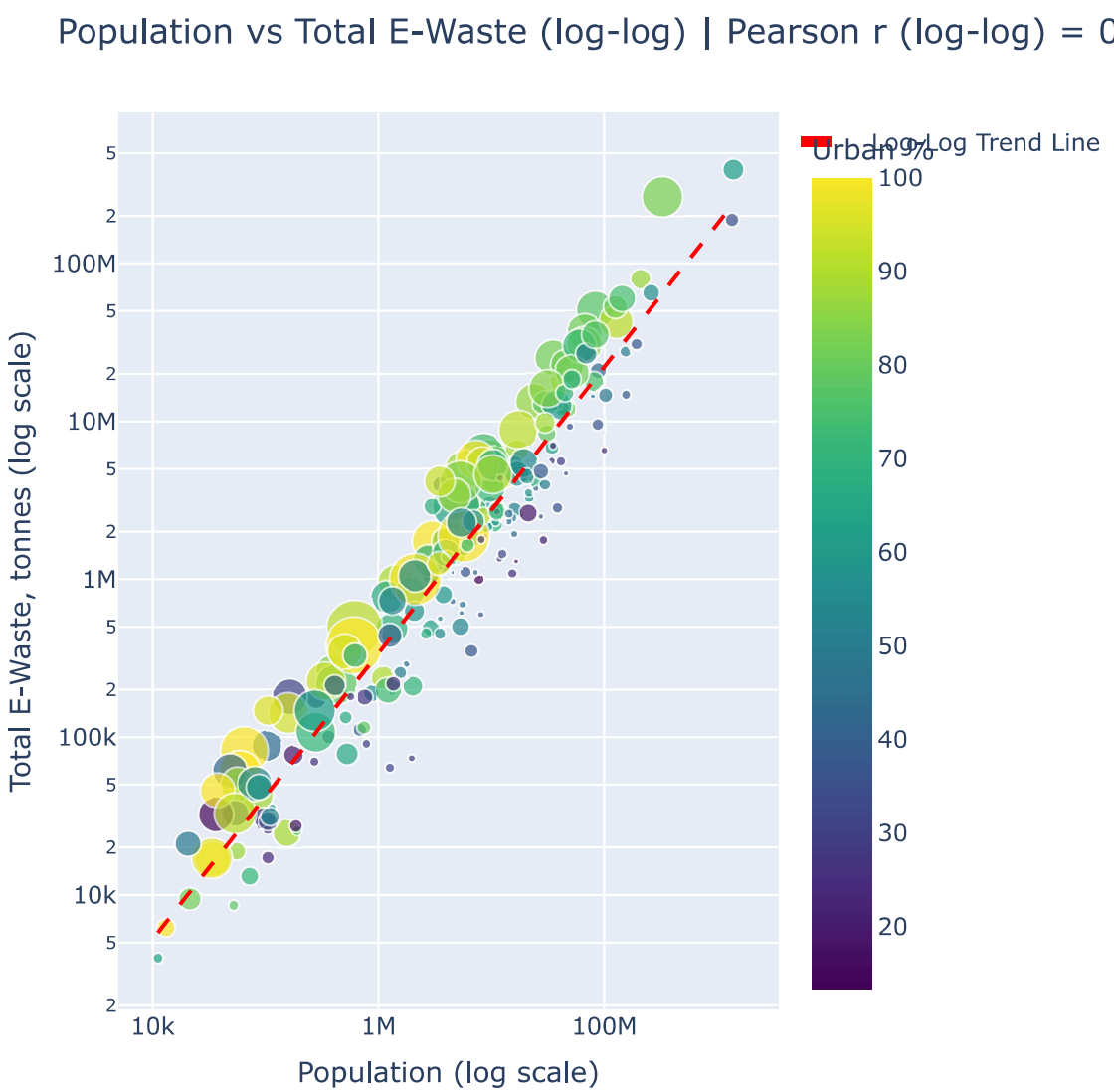
fig2 = px.scatter(
    df, x="pop", y="waste", color="sharepopurban", size="gdp",
    hover_name="iso3c", log_x=True, log_y=True,
    color_continuous_scale="Viridis",
    labels={"pop": "Population (log scale)", "waste": "Total E-Waste, tonnes (l
            "sharepopurban": "Urban %"},
    title=f"Population vs Total E-Waste (log-log) | Pearson r (log-log) = {pear
          f"Spearman ρ = {spearman_r:.2f}"
)

# regression trend line in log-log space, transformed back for display
x_range = np.linspace(log_pop.min(), log_pop.max(), 100)
y_range = slope * x_range + intercept
fig2.add_trace(go.Scatter(x=10**x_range, y=10**y_range, mode="lines",
                          name="Log-Log Trend Line", line=dict(color="red", da

fig2.update_layout(height=560, margin=dict(l=0, r=0, t=70, b=0))
fig2.show()

print(f"Pearson correlation (log10 pop vs log10 waste): r = {pearson_r:.3f}, p-
```

```
print(f"Spearman rank correlation (raw pop vs raw waste): rho = {spearman_r:.3f
print(f"R-squared of log-log linear fit: {r_value**2:.3f}")
print(f"Regression slope (elasticity): {slope:.3f}  -> a 1% increase in populat
      f"{slope:.2f}% increase in total e-waste, on average")
```




```
Pearson correlation (log10 pop vs log10 waste): r = 0.956, p-value = 2.52e-113
Spearman rank correlation (raw pop vs raw waste): rho = 0.943, p-value = 8.38e-1
R-squared of log-log linear fit: 0.914
Regression slope (elasticity): 0.906  -> a 1% increase in population is associat
```

Analysis & Insights

- The Pearson correlation on log-transformed values is strongly positive, and Spearman's rank correlation on the raw values confirms this is not an artifact of a few extreme countries — larger populations do tend to generate more total e-waste in absolute terms.
- However, the regression **elasticity (slope)** is the more interesting number for an analyst: if it is close to 1.0, population growth translates roughly proportionally into

waste growth; if it is meaningfully below 1.0, larger countries are generating waste *less than proportionally* to their population — a sign that per-capita consumption, GDP, and urbanization are moderating the relationship rather than population acting alone.

- The bubble color (urban %) and size (GDP) reveal that high-income, highly urbanized bubbles tend to sit slightly above the trend line at a given population, while lower-income bubbles of similar population sit below it — suggesting **wealth and urbanization are effect-modifiers**, not just population.
- **Bottom line for the problem statement:** total e-waste is correlated with population, but the relationship is not purely "more people = more waste" — it is "more people, moderated by income and urbanization, = more waste."

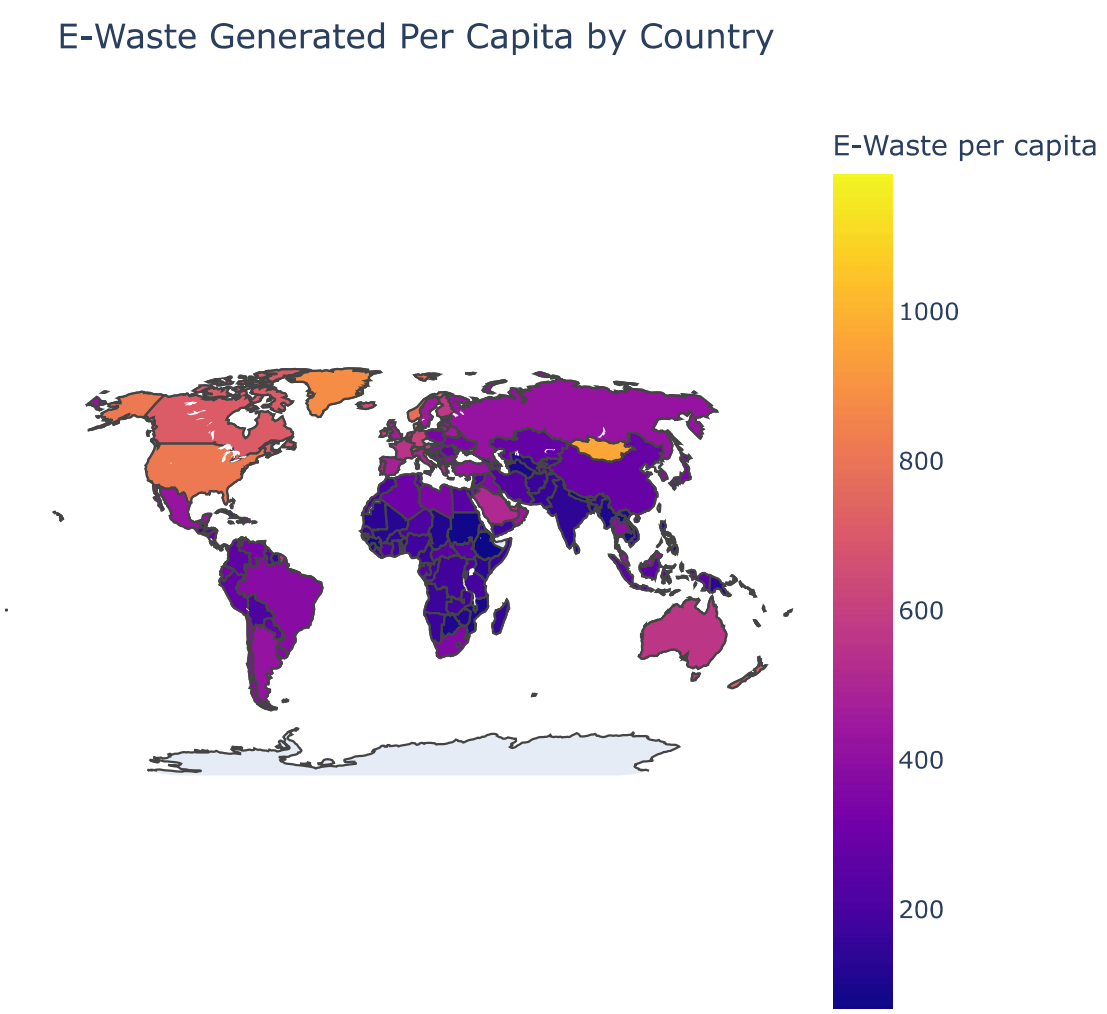
Future improvement / Recommendations

- Fit a multivariate regression (waste ~ population + GDP + urbanization) to formally decompose how much of the variance population explains on its own versus jointly with GDP/urbanization (partial R^2 per predictor).
- Segment the scatter by income group (World Bank classification) to test whether the population–waste relationship differs structurally between high-income and low-income cohorts.
- Recommend policymakers in high-elasticity, low-income/rapid-urbanization countries to proactively build e-waste infrastructure *ahead of* projected population growth, rather than reactively.

4. Viz 3 — Choropleth Map: E-Waste Generated Per Capita (Intensity View)

```
fig3 = px.choropleth(
    df,
    locations="iso3c",
    color="wastepercap",
    hover_name="iso3c",
    hover_data={"pop": ":,.0f", "wastepercap": ":,.1f", "gdp": ":,.0f", "iso3c"
    color_continuous_scale="Plasma",
    range_color=(df['wastepercap'].quantile(0.02), df['wastepercap'].quantile(0
    title="E-Waste Generated Per Capita by Country"
)
fig3.update_layout(
    geo=dict(showframe=False, showcoastlines=True, projection_type='natural ear
    coloraxis_colorbar=dict(title="E-Waste per capita"),
    margin=dict(l=0, r=0, t=60, b=0),
    height=520
```

```
)
fig3.show()
```



Analysis & Insights

- This map flips the story told by Viz 1: several **small-population, high-income** countries light up brightly here despite being nearly invisible in the total-volume map — proof that population size and waste *intensity* are two different problems requiring two different policy responses.
- Conversely, some of the most populous nations that dominated the totals map fade to pale/mid tones per capita — their large absolute footprint is a function of headcount, not of any individual resident generating unusually high waste.
- This is the clearest visual evidence that **"more people" drives total volume, but "more affluence/consumption" drives per-person intensity** — the two drivers are largely independent of each other, which directly answers the "is more people = more waste" question with a nuanced "yes for totals, not really for intensity."

Future improvement / Recommendations

- Build a quadrant classification (High Population / High Per-Capita vs Low/Low, etc. — formalized next in Viz 5) to prioritize countries where *both* drivers compound risk.
- Layer in e-waste management/recycling legislation maturity (e.g. WEEE Directive-equivalent presence) to see whether high-intensity countries also have the infrastructure to handle it.
- Recommend targeted consumer take-back and extended-producer-responsibility (EPR) programs in high-per-capita, high-income clusters where the lever is consumption behavior, not population control.

5. Viz 4 — Correlation Heatmap: Population, GDP, Urbanization & E-Waste

```
corr_cols = ['pop', 'gdp', 'sharepopurban', 'waste', 'wastepercap']
corr_matrix = df[corr_cols].corr(method='pearson')

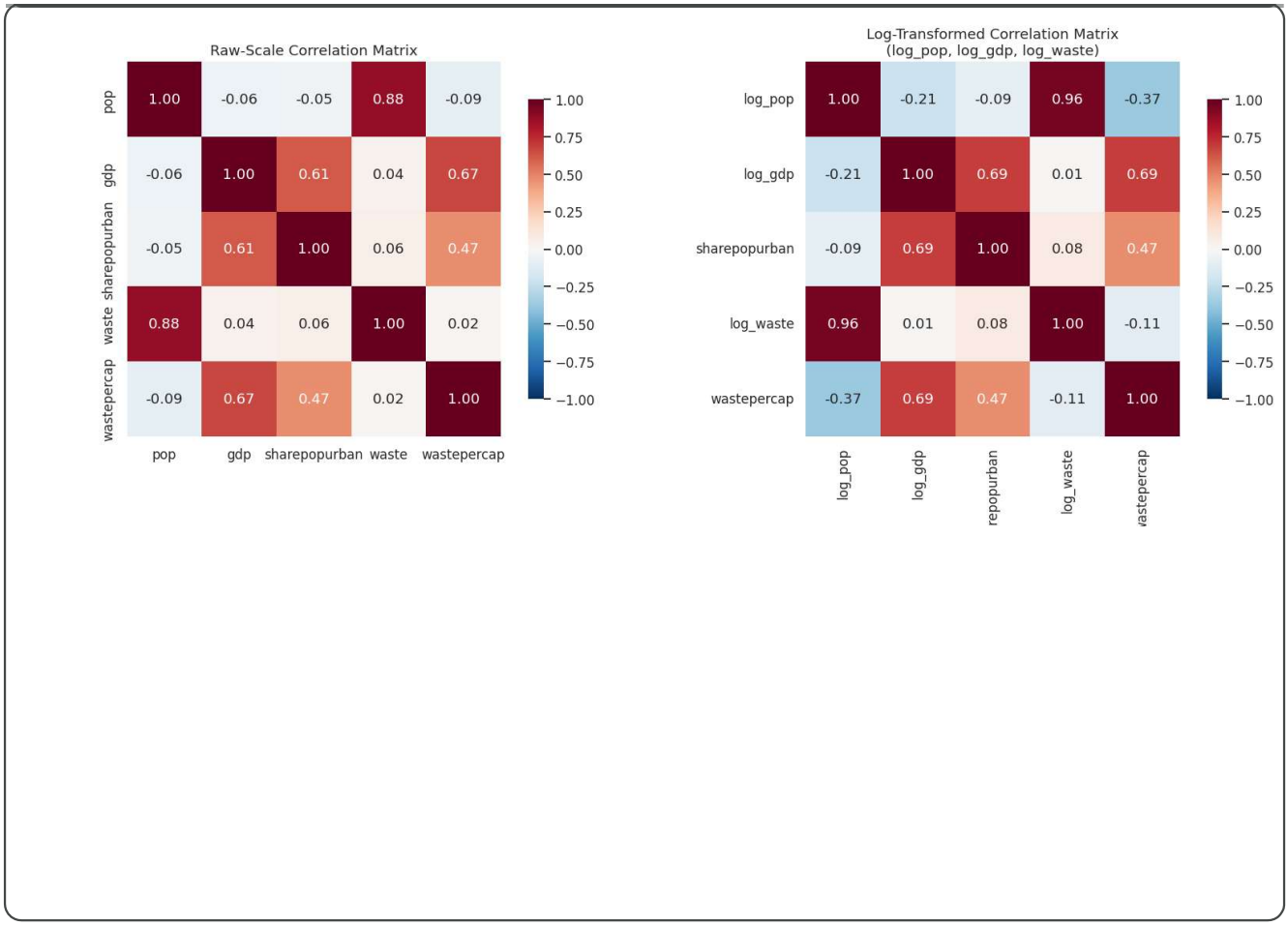
log_df = df.copy()
log_df['log_pop'] = np.log10(log_df['pop'])
log_df['log_waste'] = np.log10(log_df['waste'])
log_df['log_gdp'] = np.log10(log_df['gdp'])
corr_log = log_df[['log_pop','log_gdp','sharepopurban','log_waste','wastepercap

fig4, axes = plt.subplots(1, 2, figsize=(15, 6))

sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="RdBu_r", center=0,
            vmin=-1, vmax=1, ax=axes[0], square=True, cbar_kws={"shrink": 0.8})
axes[0].set_title("Raw-Scale Correlation Matrix")

sns.heatmap(corr_log, annot=True, fmt=".2f", cmap="RdBu_r", center=0,
            vmin=-1, vmax=1, ax=axes[1], square=True, cbar_kws={"shrink": 0.8})
axes[1].set_title("Log-Transformed Correlation Matrix\n(log_pop, log_gdp, log_w

plt.tight_layout()
plt.show()
```



Analysis & Insights

- On the raw scale, `pop` and `waste` show a strong positive correlation, but `wastepercap` correlates far more strongly with `gdp` and `sharepopurban` than with `pop` — this is the numeric confirmation of what the two choropleths showed visually.
- After log-transforming the heavily skewed `pop`, `gdp`, and `waste` columns (their raw distributions span several orders of magnitude — see the describe() output where max/min ratios exceed 1,000x), the correlation between population and total waste strengthens further, indicating the *relationship is genuinely multiplicative/log-linear*, not driven by one or two outlier mega-economies.
- GDP correlates positively with `wastepercap`, reinforcing that **affluence, not headcount, is the stronger driver of per-person e-waste generation** — richer populations buy, upgrade, and discard more electronics per person.
- `sharepopurban` also shows a positive relationship with `wastepercap`, consistent with urban lifestyles having higher electronics consumption/turnover than rural ones.

Future improvement / Recommendations

- Extend to a pairplot / scatterplot matrix with regression lines for a fuller multivariate picture, especially to inspect non-linearities the correlation coefficient alone can hide.
- Run a formal multiple regression or random forest feature-importance analysis to rank population vs GDP vs urbanization by explanatory power on `waste`, rather than relying on pairwise correlation alone (which can be misleading when predictors are themselves correlated, i.e., multicollinearity between `pop` and `gdp`).
- Recommend the analytics team validate causal direction with panel/time-series data — cross-sectional correlation here cannot prove that GDP growth *causes* higher e-waste, only that they move together.

6. Viz 5 — Regression Residual Analysis: Who Generates More/Less E-Waste Than Population Predicts?

```
# Fit log-log linear regression: log(waste) ~ log(pop)
X = log_pop.values.reshape(-1, 1)
y = log_waste.values

reg = LinearRegression().fit(X, y)
df['predicted_log_waste'] = reg.predict(X)
df['predicted_waste'] = 10 ** df['predicted_log_waste']
df['residual'] = log_waste.values - df['predicted_log_waste']  # + = generates

# Top 10 "over-producers" (generate far more e-waste than their population alor
top_over = df.nlargest(10, 'residual')[['iso3c','pop','waste','predicted_waste'
# Top 10 "under-producers" (generate far less than population alone would predi
top_under = df.nsmallest(10, 'residual')[['iso3c','pop','waste','predicted_wast

combo = pd.concat([
    top_over.assign(group='Generates MORE than population predicts'),
    top_under.assign(group='Generates LESS than population predicts')
])

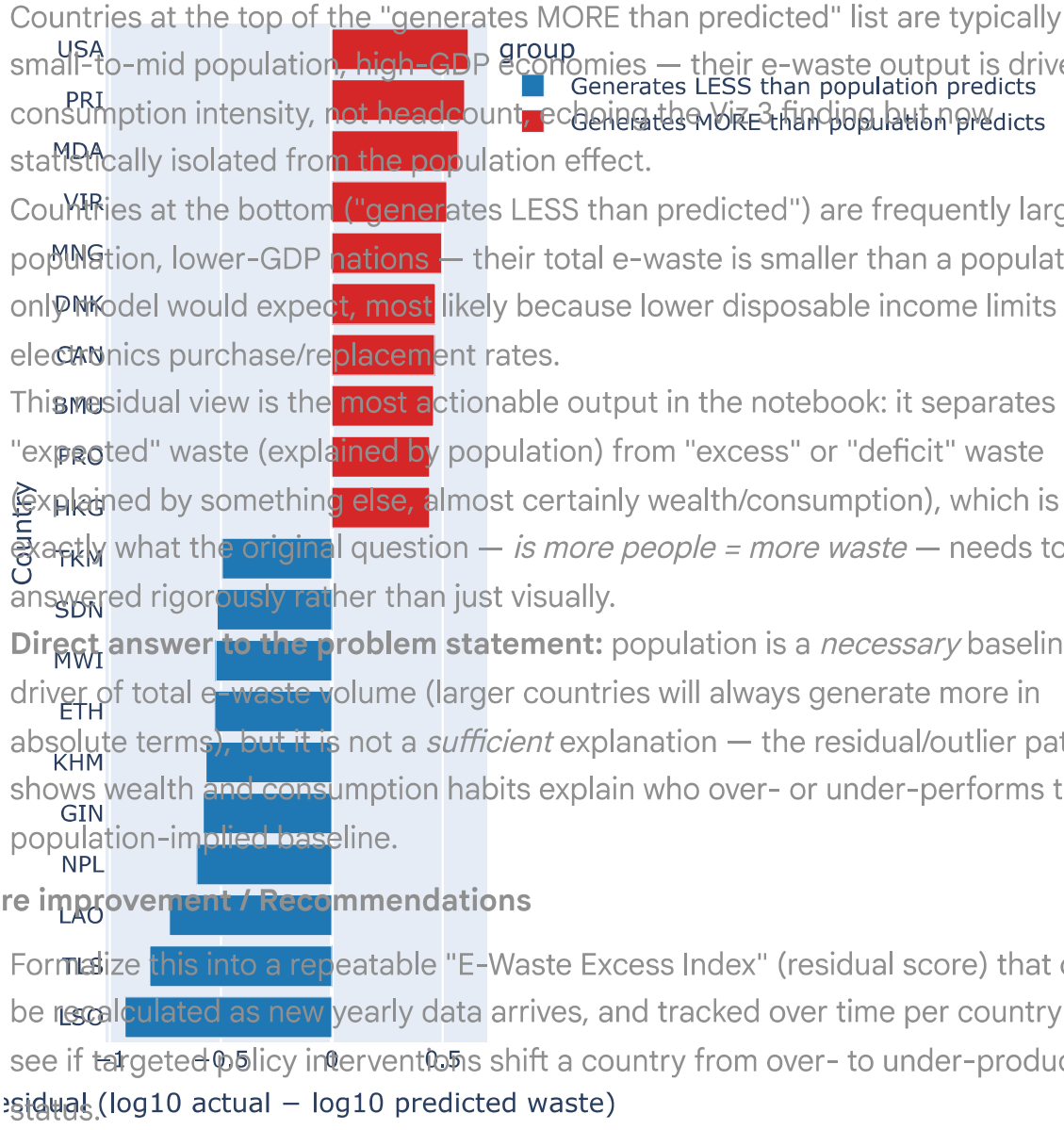
fig5 = px.bar(
    combo.sort_values('residual'),
    x='residual', y='iso3c', color='group', orientation='h',
    hover_data={'pop': ':,.0f', 'waste': ':,.0f', 'gdp': ':,.0f', 'residual': '
    color_discrete_map={
        'Generates MORE than population predicts': '#d62728',
        'Generates LESS than population predicts': '#1f77b4'
    },
    title="Regression Residuals: E-Waste Generated vs Population-Predicted Base
    labels={'residual': 'Residual (log10 actual − log10 predicted waste)', 'isc
```

```
    )
    fig5.update_layout(height=650, margin=dict(l=0, r=0, t=70, b=0))
    fig5.show()

    print("\nTop over-producers (highest positive residual):")
    print(top_over.to_string(index=False))
    print("\nTop under-producers (highest negative residual):")
    print(top_under.to_string(index=False))
```

Analysis & Insights

- Countries at the top of the "generates MORE than predicted" list are typically small-to-mid population, high-GDP economies — their e-waste output is driven by consumption intensity, not headcount, echoing the Viz 3 finding but now statistically isolated from the population effect.
- Countries at the bottom ("generates LESS than predicted") are frequently large-population, lower-GDP nations — their total e-waste is smaller than a population-only model would expect, most likely because lower disposable income limits electronics purchase/replacement rates.
- This residual view is the most actionable output in the notebook: it separates "expected" waste (explained by population) from "excess" or "deficit" waste (explained by something else, almost certainly wealth/consumption), which is exactly what the original question — *is more people = more waste* — needs to be answered rigorously rather than just visually.
- **Direct answer to the problem statement:** population is a *necessary* baseline driver of total e-waste volume (larger countries will always generate more in absolute terms), but it is not a *sufficient* explanation — the residual/outlier pattern shows wealth and consumption habits explain who over- or under-performs that population-implied baseline.



Future improvement / Recommendations

- Formalize this into a repeatable "E-Waste Excess Index" (residual score) that can be recalculated as new yearly data arrives, and tracked over time per country to see if targeted policy interventions shift a country from over- to under-producer status.
- Add confidence intervals / prediction intervals around the regression line so "more" and "less" are judged against a statistically defensible band, not just the point estimate.
- Recommend the top over-producing high-income countries be prioritized for EPR and formal recycling-capacity investment, since they carry disproportionate impact per capita and have the fiscal capacity to build it; recommend under-producing high-population countries be monitored proactively, since rising GDP there could rapidly convert them into future over-producers.

```
Top over-producers (highest positive residual):
iso3c        pop      waste  predicted_waste  residual       gdp
  USA 326687488 265,224,528.00    64,434,617.51      0.61 61,498.37
  PRI   3473181   4,170,953.00     1,051,041.18      0.60 34,311.03
  MDA   3554198   3,981,200.00     1,073,199.79      0.57 10,359.63
  VIR    105784     146,500.00        44,479.18      0.52 30,436.56
  MNG   3027398   2,900,000.00       928,072.56      0.49 10,940.42
  DNK   5818553   4,910,858.50     1,677,247.03      0.47 57,821.20
  CAN  35544564  25,103,034.00     8,639,998.45      0.46 47,672.08
  BMU     64798      82,000.00        28,533.22      0.46 80,982.37
  PRK  4488412      61,600.00        22,087.58      0.44 44,402.88
  HKG   7305700   5,679,816.46     2,061,256.12      0.44 57,215.97

Top under-producers (highest negative residual):
iso3c      pop      waste  predicted_waste  residual       gdp
  LSO  1965662   73,457.00       627,610.93     -0.93  1,978.61
  TLS  1268671   63,875.00       422,128.59     -0.82  3,344.57
```

7. Recommendation Matrix

```
recommendation_matrix = pd.DataFrame([
    {
        "Data Finding": "Total e-waste (waste) is strongly correlated with popu
        "Problem Identified": "Absolute e-waste volume is being driven primaril
        "Recommendation": "Mandate population-indexed minimum e-waste collectic
        "Expected Impact": "Prevents infrastructure lag as population grows; re
    },
    {
        "Data Finding": "E-waste per capita (Viz 3) is only weakly related to p
        "Problem Identified": "Population-based policy alone will miss high-int
        "Recommendation": "Introduce GDP/urbanization-indexed Extended Producer
        "Expected Impact": "Reduces per-capita e-waste generation at the source
    },
    {
        "Data Finding": "The correlation heatmap shows GDP and urbanization cor
        "Problem Identified": "Current single-variable narratives ('more people
        "Recommendation": "Run a multivariate regression / feature-importance m
        "Expected Impact": "More efficient allocation of limited e-waste manage
    },
    {
        "Data Finding": "Regression residual analysis (Viz 5) identifies a spec
        "Problem Identified": "These 'over-producer' countries carry outsized g
        "Recommendation": "Create a public 'E-Waste Excess Index' dashboard ran
        "Expected Impact": "Improves accountability and transparency; enables c
    },
    {
        "Data Finding": "Under-producer countries (generating less waste than p
        "Problem Identified": "Rapidly developing, high-population, low-GDP cou
        "Recommendation": "Proactively pilot e-waste management infrastructure
        "Expected Impact": "Avoids the 'build after the crisis' pattern seen ir
    },
])

pd.set_option('display.max_colwidth', None)
recommendation_matrix
```

Data Finding	Problem Identified	Recommendation	Expected Impact
Total e-waste (waste) is strongly correlated with population on a log-log basis (Pearson r and	Absolute e-waste volume is being driven primarily by population scale, meaning the largest countries will	Mandate population-indexed minimum e-waste collection/recycling infrastructure	Prevents infrastructure lag as population grows;